← Back **Practice Quiz: Flood Fill**
Practice Assignment • 30 min

English

Your grade: 100%

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Next item →

1. In the context of pathfinding, the Flood Fill algorithm can be utilized as a foundational technique for mapping out reachable areas within a grid or maze from a given starting point. This can be particularly useful in scenarios where the goal is to identify all possible destinations or to preprocess the map for further pathfinding computations. What is the primary purpose of using Flood Fill in the context of pathfinding at a conceptual level?

1 / 1 point

- ○ To compress the map data by identifying and eliminating redundant paths
- ○ To enhance the graphical representation of the map for better visualization
- ◉ To identify and fill all areas within a map that are reachable from a given starting point, marking the extent of accessible territory
- ○ To calculate the shortest path between two points on the map
- ○ To dynamically alter the terrain of the map based on the movement of entities

Correct
Correct. In pathfinding, the primary purpose of Flood Fill is to algorithmically explore and mark all locations that can be reached from a starting point, effectively delineating the extent of accessible territory. This can be crucial for understanding the navigable areas within a map before applying more specific pathfinding algorithms to determine optimal routes.

2. In the implementation of the Flood Fill algorithm for pathfinding or area identification in a grid, the propagate function plays a crucial role in expanding the search to cover all accessible areas from a given starting point. Conceptually, what does the propagate function achieve within the Flood Fill algorithm?

1 / 1 point

- ○ The propagate function compresses the grid to optimize storage
- ○ The propagate function renders the visual representation of the flood fill on the screen
- ◉ The propagate function checks neighboring cells and expands the search to allow for the flood fill area to grow
- ○ The propagate function calculates the shortest path from the start point to all other points in the grid
- ○ The propagate function dynamically changes the grid's layout based on the entities' movements

Correct
Correct. This description accurately captures the essence of the propagate function within the Flood Fill algorithm. By checking neighboring cells and recursively or iteratively expanding the search, the propagate function enables the algorithm to systematically explore and mark all reachable areas from the initial starting cell, effectively growing the flood-filled region.

3. In applications of the Flood Fill algorithm, such as mapping out accessible areas in a grid for pathfinding or area identification, encountering obstacles is a common challenge. These obstacles are areas within the grid that cannot be traversed or filled. How does the Flood Fill algorithm detect obstacles when propagating through the grid to ensure that the fill does not extend beyond designated boundaries?

1 / 1 point

- ○ By incrementing a counter for every cell it fills, stopping when a maximum count is reached
- ○ Using a global map that marks the locations of all obstacles in advance
- ◉ By checking the value or property of each neighboring cell before deciding to fill it
- ○ Randomly selecting cells to fill and skipping those that do not meet a specific random criterion
- ○ Automatically filling all cells within a predetermined radius, ignoring any potential obstacles

Correct
Correct. The Flood Fill algorithm detects obstacles by evaluating the properties (such as color, value, or a flag indicating an obstacle) of each neighboring cell before deciding whether to propagate into that cell. If a neighboring cell meets the criteria for being filled (e.g., it has the same color as the target area and is not marked as an obstacle), the algorithm proceeds to fill it; otherwise, the cell is considered an obstacle, and the fill does not extend into it.